

Multiple Choice Questions: Ratio and Proportions

Mathematics Practice Module

December 30, 2025

1 Practice Questions

- Q1.** If $A : B = 2 : 3$ and $B : C = 4 : 5$, then $A : C$ is:
A) 8:15 B) 2:5 C) 6:15 D) 8:12
- Q2.** The fourth proportional to 5, 8, 15 is:
A) 18 B) 24 C) 19 D) 20
- Q3.** If $x : y = 3 : 4$, then $(7x + 3y) : (7x - 3y)$ is:
A) 5:2 B) 4:3 C) 11:3 D) 37:19
- Q4.** The mean proportional between 9 and 25 is:
A) 17 B) 15 C) 225 D) 16
- Q5.** If 40% of a number is equal to two-third of another number, what is the ratio of the first number to the second number?
A) 2:5 B) 3:7 C) 5:3 D) 7:3
- Q6.** Two numbers are in the ratio 3:5. If 9 is subtracted from each, the new numbers are in the ratio 12:23. The smaller number is:
A) 27 B) 33 C) 49 D) 55
- Q7.** If $A : B = 5 : 7$ and $B : C = 6 : 11$, then $A : B : C$ is:
A) 30:42:77 B) 30:42:66 C) 5:42:11 D) 30:7:11
- Q8.** A sum of money is to be distributed among A, B, C, D in the proportion of 5:2:4:3. If C gets \$1000 more than D, what is B's share?
A) \$500 B) \$1500 C) \$2000 D) \$2500
- Q9.** The ratio of the number of boys and girls in a college is 7:8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?
A) 8:9 B) 21:22 C) 17:18 D) Cannot be determined
- Q10.** In a mixture of 60 liters, the ratio of milk and water is 2:1. If this ratio is to be 1:2, then the quantity of water to be further added is:
A) 20 liters B) 30 liters C) 40 liters D) 60 liters
- Q11.** If $x : y = 5 : 2$, then $(8x + 9y) : (8x + 2y)$ is:
A) 22:29 B) 26:61 C) 29:22 D) 61:26

- Q12.** The third proportional to 12 and 18 is:
 A) 3 B) 6 C) 27 D) 144
- Q13.** If \$510 be divided among A, B, C in such a way that A gets $\frac{2}{3}$ of what B gets and B gets $\frac{1}{4}$ of what C gets, then their shares are respectively:
 A) \$60, \$90, \$360 B) \$50, \$100, \$360 C) \$40, \$120, \$350 D) \$150, \$300, \$60
- Q14.** If $2A = 3B = 4C$, then $A : B : C$ is:
 A) 2:3:4 B) 4:3:2 C) 6:4:3 D) 3:4:6
- Q15.** A bag contains 50p, 25p and 10p coins in the ratio 5:9:4, amounting to \$206. Find the number of coins of each type.
 A) 200, 360, 160 B) 160, 360, 200 C) 200, 160, 300 D) 300, 200, 160
- Q16.** If $x/y = 6/5$, then the value of $(x^2 + y^2)/(x^2 - y^2)$ is:
 A) 61/11 B) 11/61 C) 36/25 D) 25/36
- Q17.** The ratio of three numbers is 3:4:7 and their product is 18144. The numbers are:
 A) 9, 12, 21 B) 15, 20, 35 C) 18, 24, 42 D) 21, 28, 49
- Q18.** If $a : b = c : d$, then $(ma + nc)/(mb + nd)$ is equal to:
 A) a/b B) c/d C) Both A and B D) None of these
- Q19.** Zinc and Copper are in the ratio 5:3 in 400g of an alloy. How much of Copper (in grams) should be added to make the ratio 5:4?
 A) 50g B) 66g C) 72g D) 200g
- Q20.** A fraction which bears the same ratio to $1/27$ as $3/11$ does to $5/9$ is:
 A) $1/55$ B) 55 C) $1/11$ D) $3/11$
- Q21.** If x is subtracted from each of 6, 7, 15 and 17, the numbers so obtained are in proportion. The value of x is:
 A) 1 B) 2 C) 3 D) 4
- Q22.** If $x : y = 2 : 1$, then $(x^2 - y^2) : (x^2 + y^2)$ is:
 A) 3:5 B) 5:3 C) 1:3 D) 3:1
- Q23.** The salary of A, B and C are in the ratio 2:3:5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be the new ratio of their salaries?
 A) 3:3:10 B) 10:11:20 C) 23:33:60 D) Cannot be determined
- Q24.** What is the duplicate ratio of 3:4?
 A) $\sqrt{3} : \sqrt{4}$ B) 9:16 C) 27:64 D) 4:3
- Q25.** In a library, the ratio of story books to other books is 7:2 and there are 1512 story books. Due to collection of some more story books, the said ratio becomes 15:4. The number of story books collected is:
 A) 84 B) 100 C) 108 D) 205
- Q26.** The sub-triplicate ratio of 8:27 is:
 A) 2:3 B) 4:9 C) 64:729 D) 512:19683

- Q27.** If $(a + b) : (a - b) = 15 : 1$, then $a^2 - b^2$ is proportional to:
A) ab B) $a^2 + b^2$ C) a/b D) All of these
- Q28.** A and B together have \$1210. If $4/15$ of A's amount is equal to $2/5$ of B's amount, how much amount does B have?
A) \$460 B) \$484 C) \$550 D) \$664
- Q29.** If $(x + y) : (x - y) = 4 : 1$, then $(x^2 + y^2) : (x^2 - y^2)$ is:
A) 25:16 B) 16:25 C) 17:8 D) 8:17
- Q30.** Two numbers are in the ratio 1:2. If 7 is added to both, their ratio changes to 3:5. The greatest number is:
A) 24 B) 26 C) 28 D) 32

2 Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	7	A	13	A	19	A	25	C
2	B	8	C	14	C	20	A	26	A
3	C	9	B	15	A	21	C	27	A
4	B	10	D	16	A	22	A	28	B
5	C	11	C	17	C	23	C	29	C
6	B	12	C	18	C	24	B	30	C

3 Core Concepts

- **Proportion:** If a, b, c, d are in proportion, then $a/b = c/d$ or $ad = bc$.
- **Mean Proportional:** The mean proportional between a and b is $\sqrt{ab}$.
- **Third Proportional:** The third proportional to a and b is x such that $a : b = b : x \implies x = b^2/a$.
- **Compounded Ratio:** The compounded ratio of $(a : b)$ and $(c : d)$ is $(ac : bd)$.
- **Duplicate Ratio:** The duplicate ratio of $(a : b)$ is $(a^2 : b^2)$.
- **Componendo and Dividendo:** If $a/b = c/d$, then $(a + b)/(a - b) = (c + d)/(c - d)$.